MCQ

Question 1 (2 points)

Consider the following languages:

$$M_1 = \{a^n b^m c^n \mid m, n \geq 0\}, \quad M_2 = \{a^n b^n c^m \mid m, n \geq 0\}$$

Which of the following languages is not context-free?

A. M_1

B. M_2

C. $M_1 \cap M_2$

D. $M_1 \cup M_2$

Option C is correct.

M_1 and M_2 are both context free languages. We can easily construct CFGs for both these languages. Union of two context free languages is also context-free.

Now, $M_1 \cap M_2 = \{a^n b^n a^n c^n \mid n \geq 0\}$. We can show using pumping lemma for context free languages that $M_1 \cap M_2$ is not context free.

Question 2 (2 points)

Which of the following classes is exactly the set of languages accepted by deterministic Push Down Automata?

A. All context-free languages

B. Deterministic context-free languages

C. All regular languages

D. Context-sensitive languages

Option B is correct.

The class of languages accepted by deterministic push down automata is Deterministic Context Free Languages.

Question 3 (2 points)

Which of the following languages is context-free?

A. $\{0, 1\}^*$

B. $\{a^n b^n c^n \mid n \geq 0\}$

C. $\{ww^R w \mid w \in \{0, 1\}^*\}$

D. $\{0^p \mid p \text{ is prime}\}$

Option A is correct.

Using the pumping lemma for context free languages, we can show that options B, C and D are not context-free languages. Option A is the universal language, which is trivially a context-free language.

Question 4 (2 points)

Which of the following languages is not context-free?

A. $\{a^n b^m c^n d^m \mid n \geq 0\}$

B. $\{a^i b^j a^i \mid i, j \geq 0\}$

C. Set of binary strings with no substring 1010

D. Set of strings over $\{a, b\}$ where the number of a's is even

Option A is correct.

Options C and D are regular languages and hence also context-free. Option B is also context-free. We can easily construct a CFG for B.

We can show using pumping lemma for context-free languages that option A is not context-free.

Question 5 (2 points)
Which of the following is a distinguishing feature of a pushdown automaton?

A. Multiple random-access work tapes

B. A single stack

C. A circular buffer

D. A priority queue

Option B is correct.

A push down automata contains only a single stack.

Question 6 (2 points)
Consider the following:

$$P_1 = \{0^n 0^n 1^n \mid n \geq 1\}, \quad P_2 = \{0^n 1^n 0^n \mid n \geq 1\}$$

Which of the following is TRUE?

A. Both P_1 and P_2 are context-free

B. P_1 is context-free, P_2 is not

C. P_2 is context-free, P_1 is not

D. Neither P_1 nor P_2 is context-free

Option B is correct.

We can construct a CFG for P_1 as follows:

$$S \to 00S1 \mid \epsilon$$

. We can show using the pumping lemma or context free languages that P_2 is not context-free.

Question 7 (2 points)
Let $Q = \{w \in \{x, y\}^* \mid w$ has equal number of x and $y\}$. What can be said about Q?

A. Q can be accepted by a DFA

B. Q cannot be accepted by an NFA but can be accepted by a deterministic PDA

C. Q cannot be accepted by a deterministic PDA but can be accepted by a nondeterministic PDA

D. Q cannot be accepted by a DFA but can be accepted by an NFA

Option B is correct.

Q is a deterministic context-free language but not a regular language (can be shown using pumping lemma for regular languages). A deterministic PDA can be constructed for Q as follows: Push x to stack if the stack top is x or empty. Similarly, push y to stack if the stack top is y or empty. If the stack top is x and the next bit in the input string is y, then pop x from the stack and vice-versa. Accept the string if the stack is empty after processing the entire input string. Hence,Q can be accepted by a deterministic PDA but not by DFA or NFA.

Question 8 (2 points)
Which of the following statements is true?

A. PDA acceptance by empty stack is more powerful than acceptance by final state

B. PDA acceptance by empty stack is less powerful than acceptance by final state

C. PDA acceptance by empty stack is equally powerful as acceptance by final state

D. PDA acceptance by empty stack and by final state are incomparable

Option C is correct.

PDA accepting by final state and PDA accepting by empty stack are equivalent in power.